

AFA3732A-R1 Series



1. Features of AFA3732A-R1 Series :

- Immersion cooling ready power beads.
- Inductance Range: 40.0nH to 300.0nH, Custom values are welcomed.
- High current output chokes, up to 210.0 Amp with approx. 20% roll off.
- Low Profile 8.00 mm Max. height .
- Foot Print 9.60 x 6.40 mm Max .
- Perfect for high density designs with limited board space.
- Operating frequency up to 5.0 MHz application.
- Operating Temperature Range -55° C to +130° C , RoHs & HF compliance .
- T & R Qty: 600 pcs , 13" Reel.

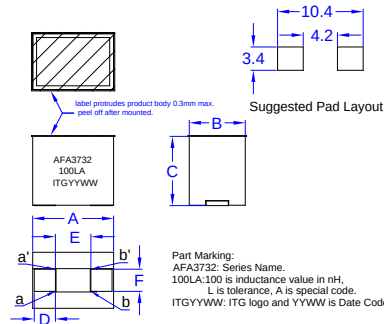


2. Electrical Characteristic of AFA3732A-R1 Series:

ITG Part Number	OCL ¹ (nH) ± 15%	L @ Isat ² (nH) Min.	DCR ³ (mΩ) ± 5%	Isat1 ⁴ (A) @25°C	Isat2 ⁴ (A) @75°C	Isat3 ⁴ (A) @100°C	Irms ⁵ (A) @25°C
AFA3732A-40L-R1	40.00	28.80	0.145	210.00	190.00	180.00	75.00
AFA3732A-50L-R1	50.00	36.00	0.145	185.00	175.00	160.00	75.00
AFA3732A-60L-R1	60.00	43.20	0.145	155.00	150.00	140.00	75.00
AFA3732A-70L-R1	70.00	50.40	0.145	124.00	120.00	115.00	75.00
AFA3732A-80L-R1	80.00	57.60	0.145	118.00	110.00	102.00	75.00
AFA3732A-100L-R1	100.00	72.00	0.145	95.00	88.00	82.00	75.00
AFA3732A-120L-R1	120.00	86.40	0.145	78.00	71.00	67.00	75.00
AFA3732A-150L-R1	150.00	108.00	0.145	60.00	54.00	52.00	75.00
AFA3732A-180L-R1	180.00	129.60	0.145	50.00	48.00	46.00	75.00
AFA3732A-220L-R1	220.00	158.40	0.145	38.00	36.00	34.00	75.00
AFA3732A-270L-R1	270.00	194.40	0.145	30.00	28.00	26.00	75.00
AFA3732A-300L-R1	300.00	216.00	0.145	26.00	24.00	22.00	75.00

3. Mechanical Dimension(Unit : mm):

A	B	C	D	E	F
Max.	Max.	Max.	± 0.35	Nom.	± 0.20
9.60	6.40	8.00	2.35	4.70	2.80



Notes:

1. Open Circuit Inductance (OCL) test condition:500KHz,0.25Vrms,0Adc ,at 25 °C.
2. L @ Isat and L @ Irms Test condition:500KHz,0.25Vrms (Ta=25 °C).
3. The nominal DCR is measured from point "a" and "a" to point "b" and "b", as shown above on the mechanical drawing (Ta=25°C).
4. Isat1,Isat2 & Isat3 : DC current that will cause inductance to drop approximately by 20%.
5. Irms : DC current for an approximate temperature rise of 40°C without core loss , Derating is necessary for AC currents. PCB pad layout , trace thickness and width , air-flow and proximity of other heat generating components will affect the temperature rise. It is recommended the part temperature not exceed 130 °C under worst case operating conditions verified in the end application.
6. Please contact ITG Sales or FAE to discuss your immersion cooling solvent and manufacturing process, ITG engineers will verify each customer's immersion cooling PCBA process.



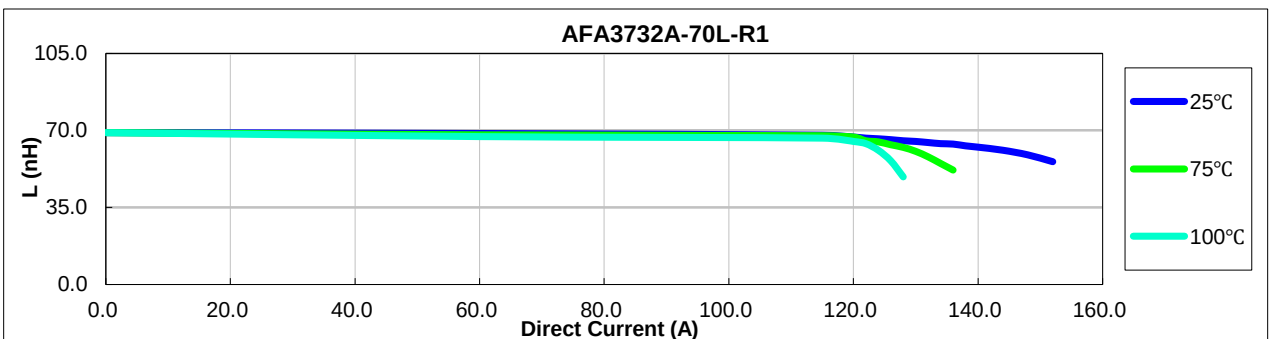
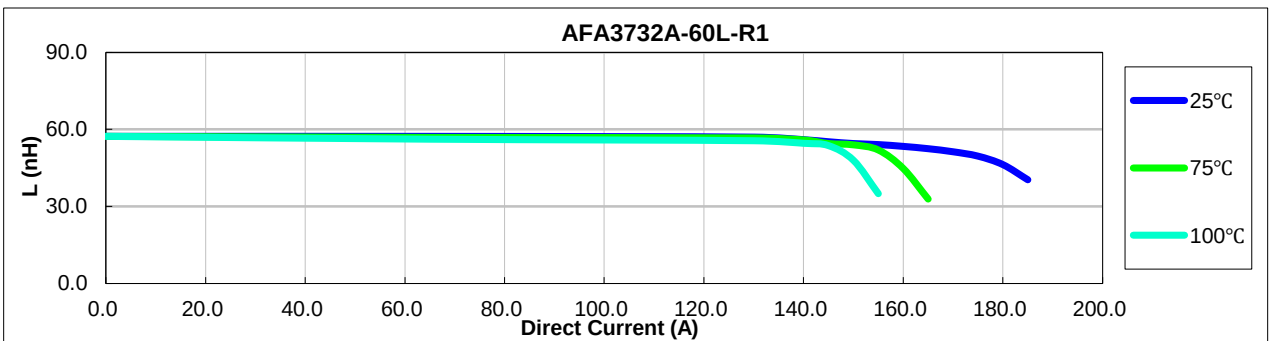
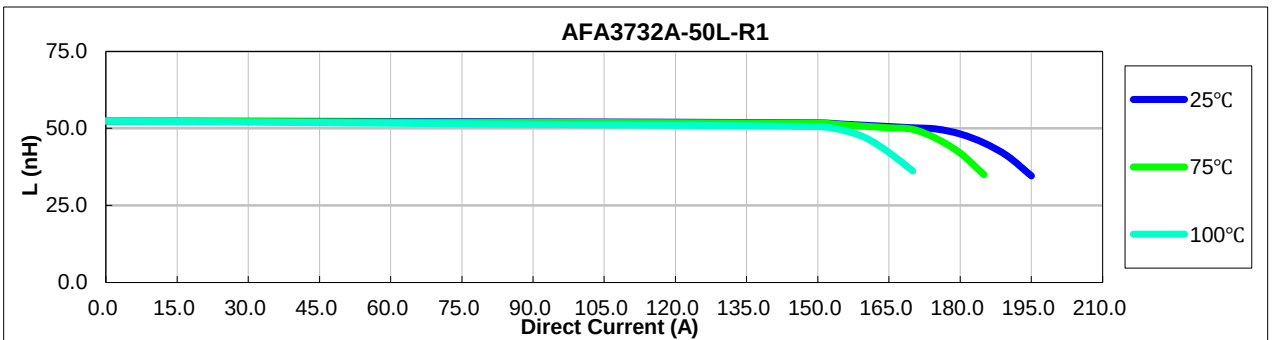
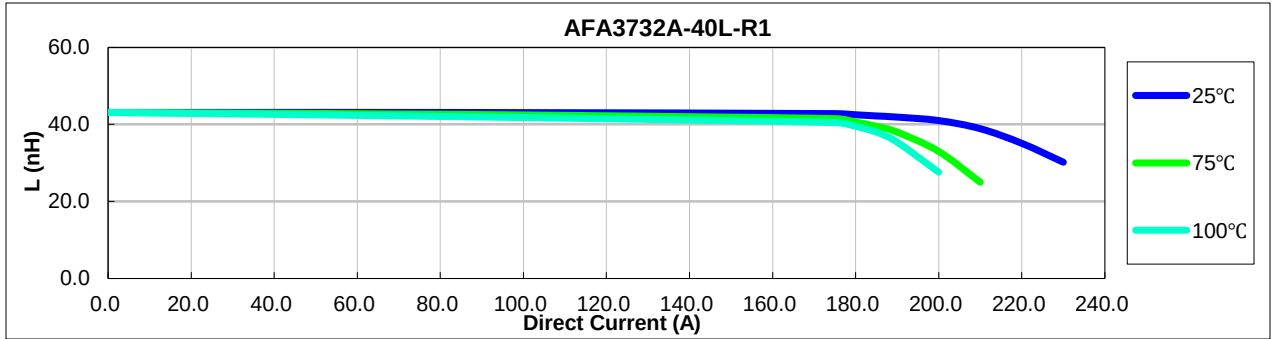
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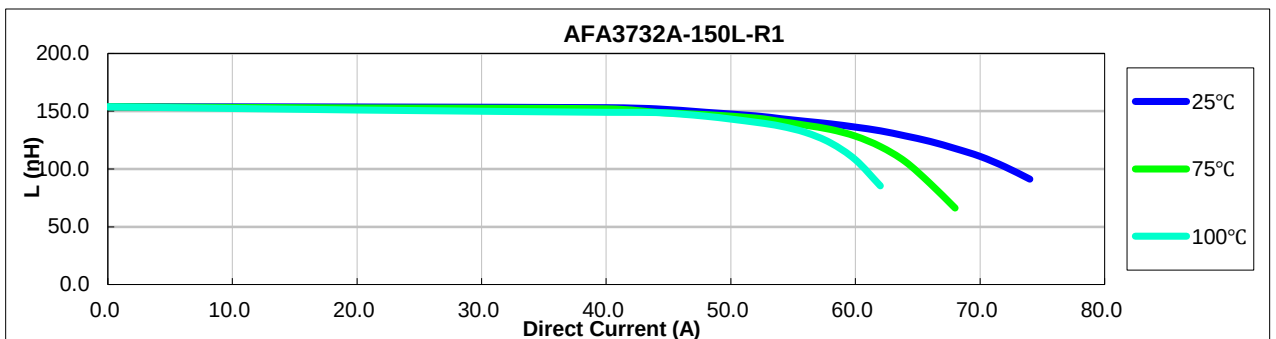
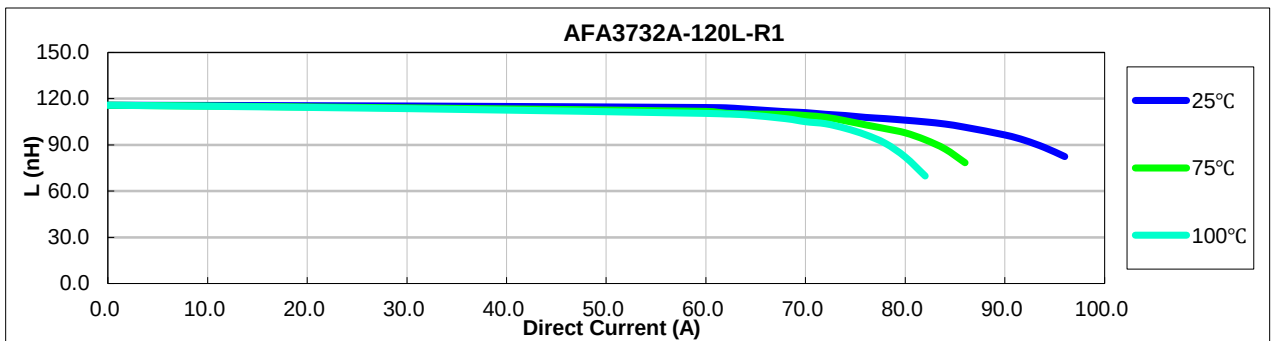
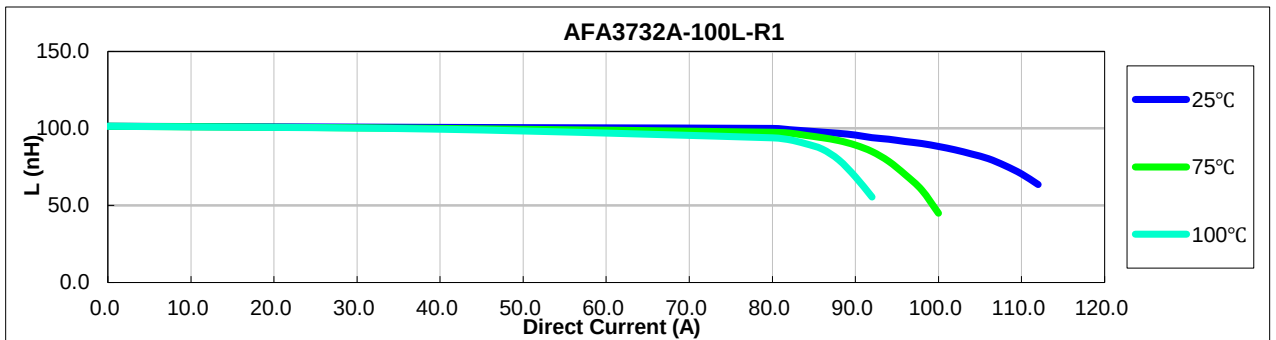
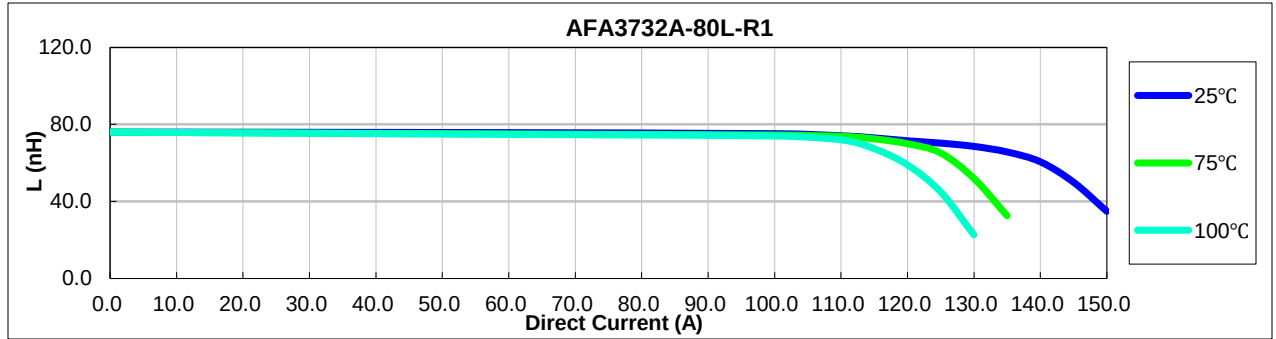
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*Due to continuous product improvement, all specifications are subject to change without prior notice. Kindly contact an ITG field application engineer or a sales representative prior to purchase.

4. Inductance Characteristics (Inductance vs. Current):



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